Rudder – Maintenance Practices

1. Removal/Installation

**CAUTION** **IF REMOVING OR INSTALLING CONTROL CABLES, DISCONNECT MAIN BATTERY (REFER TO 24-30-31, MAIN BATTERY SYSTEM – MAINTENANCE PRACTICES, PARA 2.), AVIONICS START BATTERY (IF INSTALLED, REFER TO 24-30-32, AVIONICS START BATTERY (MOD 6/2315) – GENERAL DATA, PARA 4.) AND EMERGENCY BATTERY (IF INSTALLED, REFER TO 33-50-01, EMERGENCY LIGHTING BATTERY – GENERAL DATA, PARA 1.) AND DISCONNECT GROUND POWER.**

A. Remove Rudder

1) Remove fairings at bottom of vertical stabilizer.

2) Disconnect electrical wiring to tail light, at quick disconnect (refer to 33-00-00, Lights – General Data) and sensor wiring at WL 50.15 FWD of rudder hinge line.

3) Slacken and disconnect rudder control cables at rudder lever and slacken rudder trim cables at turnbuckles in rear fuselage.

4) Release rudder trim cables from screw jack.

5) Disconnect geared tab control rod and link (refer to 27-26-11, Rudder Geared Tab – Maintenance Practices).

6) Attach sling SD12517 to rudder and take weight (refer to 07-00-00, Lifting and Shoring – General Data).

7) Support rudder and remove lower and center hinge bolts and bonding. At top hinge position, remove two bolts from upper hinge plate. Lower rudder to ground and remove upper hinge attachment bolt to remove upper hinge plate.

8) Disconnect rudder trim tab connecting rod and remove trim screw jack.

B. Install Rudder

Note If the rudder to be installed has been repaired, painted or repainted, check the balance in accordance with Structural Repair Manual PSM 1–6–3.

1) Examine replacement rudder for signs of damage.

2) Install hinge plate to rudder upper hinge using attachment bolt.

3) Install rudder trim screw jack in rudder.

4) Attach sling SD12517 to rudder (refer to 07-00-00, Lifting and Shoring – General Data).

5) Position rudder and secure to vertical stabilizer lower and center hinges, using bolts, washers and nuts.

Note Pass bolt through bonding jumper terminal before inserting in lug; use two washers when installing each bolt, one between bonding jumper terminal and lug, sensor wiring at WL 50.15 FWD of rudder hinge line, and one under nut.

6) Secure upper hinge plate to vertical stabilizer, using two bolts, washers and nuts.

7) Tighten all rudder attachment bolts.

8) Install trim tab connecting rod.

9) Connect geared tab control rod and link (refer to 27-26-11, Rudder Geared Tab – Maintenance Practices).

10) Connect rudder cables to rudder lever, and secure trim cables to trim screw jack.

11) Rig rudder control system (refer to 27-20-00, Rudder Control System – General Data).

12) Rig rudder trim tab system (refer to 27-23-00, Rudder Trim Tab System – General Data).

13) Check rudder and rudder trim tab control surface clearances (refer to 27-00-00, Flight Controls – General Data).

14) Connect tail light electrical wiring at quick disconnect (refer to 33-00-00, Lights – General Data).

15) Operate rudder and rudder trim controls and check for full and free range of movement.

16) Do an operational test for the rudder angle position indicator as per section 27-20-31, Rudder Angle Position Indicator – General Data, Para 3.B.

17) If adjustment is required for the rudder angle position potentiometer adjust as per 27-20-31, Rudder Angle Position Indicator – General Data, Para 3.A.